



**SIMATS**  
ENGINEERING



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Saveetha Institute of Medical And Technical Sciences  
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

## Exp No 2: Water Jug Problem

### Aim:

To solve the Water Jug Problem using BFS in Python.

### Algorithm

1. Read jug capacities and target.
2. Start from state (0, 0).
3. Generate all possible states.
4. Check if target is reached.
5. Print the solution path.

### Pseudocode

START

Create Queue

Mark source as visited

Insert source into Queue

WHILE Queue not empty

    Remove front node

    Print node

    FOR every adjacent node

        IF not visited

            Mark visited

            Insert into Queue

        ENDIF

    ENDFOR

ENDWHILE

STOP

### Program:

```
from collections import deque
def water_jug(a, b, target):
    q = deque([(0, 0)])
    visited = set()
```

```

while q:
    x, y = q.popleft()
    if (x, y) in visited:
        continue
    visited.add((x, y))
    print(x, y)
    if x == target or y == target:
        return
    q.append((a, y))
    q.append((x, b))
    q.append((0, y))
    q.append((x, 0))
    q.append((max(0, x-(b-y)), min(b, y+x)))
    q.append((min(a, x+y), max(0, y-(a-x))))
a = int(input())
b = int(input())
target = int(input())
water_jug(a, b, target)

```

## Output:

```

IDLE Shell 3.14.0
File Edit Shell Debug Options Window Help
Python 3.14.0 (tags/v3.14.0:ebf955d, Oct 7 2025, 10:15:03) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/bhuva/Downloads/exp 2.py =====
4
3
2
0 0
4 0
0 3
4 3
1 3
3 0
1 0
3 3
0 1
4 2
>>> |

```

## Result:

The program displays the sequence of water jug states until the target amount is reached.